

Vietnamese Emotion Recognition from Voice and Text: A Confidence-Based Approach

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Abstract—Accurate emotion recognition from human communication is pivotal for empathetic human–computer interaction, healthcare monitoring, and customer service. However, most systems either focus on speech or text, and few effectively integrate both, especially for underrepresented languages like Vietnamese. In this study, we propose a lightweight confidence-based multimodal fusion framework that combines a HuBERT-based voice classifier and a PhoBERT-v2 text classifier fine-tuned with LDAM loss and Deferred Re-Weighting (DRW). Each branch produces posterior probabilities, and the final prediction selects the modality with higher confidence, avoiding complex fusion networks and preserving real-time performance. On Vietnamese benchmarks, the text model achieves 87.35% accuracy and an F1-score of 87.28% on an extended UIT-VSMEC corpus, while the speech model attains 82.06% accuracy and an F1-score of 81.93% on the Bud500 dataset. The proposed fusion strategy shows promising results, boosting macro-F1 compared to single-modality baselines. Our results demonstrate that simple, confidence-driven fusion is highly effective for Vietnamese emotion-aware applications, offering a practical path for deployment in resource-constrained and real-time settings.

Index Terms—Emotion Recognition, Multimodal Fusion, Voice Emotion Recognition, Text Emotion Classification, PhoBERT, HuBERT.

I. INTRODUCTION

Emotion recognition is an important branch of Affective Computing, with applications in Human-Computer Interaction (HCI), Healthcare, and Customer Service. Accurately identifying and interpreting emotions plays a key role in improving user experience by allowing systems to respond empathetically and provide personalized services. In HCI, emotion recognition helps systems become more adaptive and intuitive to users' emotional states, while in healthcare, it supports patient monitoring for better diagnoses and treatments. In customer service, real-time emotion detection allows timely interventions to improve service quality [1], [2].

Voice and text are two prominent modalities in emotion recognition, each offering unique strengths. Voice conveys rich emotional information through acoustic variations such as tone, pitch, and speech rate, making it a reliable indicator of emotional states [3]. Conversely, text provides emotional insights via lexical choices, syntactic patterns, and contextual

clues, capturing subtle nuances often missed by voice-only analyses [4]. Combining these modalities holds promise for significantly enhancing accuracy and achieving comprehensive emotional understanding.

However, most existing research focuses on either voice or text, with many studies failing to integrate both modalities effectively. Early work focused on audio-based emotion recognition, while more recent studies incorporate text-based sentiment analysis. Despite the emergence of multimodal methods, many approaches do not fully leverage the complementary strengths of voice and text [5]. Furthermore, most emotion recognition systems have been designed for major languages like English, leaving languages like Vietnamese underrepresented [6]. The lack of high-quality data for Vietnamese further complicates the development of accurate models for this language.

This paper proposes a novel multimodal emotion recognition system, intelligently combining voice and text modalities to maximize accuracy and robustness. Our approach addresses existing limitations by leveraging complementary strengths, thus providing a more holistic understanding of emotions. Such multimodal integration is particularly beneficial for languages with limited resources, including Vietnamese, offering practical improvements in real-world applications. The subsequent sections discuss the background, methodologies, and specific techniques that underpin our proposed multimodal fusion approach.

II. BACKGROUND AND RELATED WORK

Multimodal emotion recognition (MER) has progressed rapidly thanks to advances in deep representation learning for both speech and text. Below we briefly review the most relevant milestones and pinpoint the gap this work fills.

A. Text-based Emotion Classification

Identifying emotions from text underpins opinion mining, mental-health screening, and empathetic dialogue systems. Public corpora such as GoEmotions (58k Reddit comments) and UIT-VSMEC (6.9k Vietnamese sentences) enable fine-grained modelling, although baseline BERT on GoEmotions

still scores only 0.46 F1, highlighting the task’s difficulty [7], [8].

Classic approaches (Naïve Bayes, SVM + TF-IDF) remain attractive on constrained hardware but exploit only surface features [9]. Recurrent networks (LSTM, GRU) capture temporal context, and CNN-LSTM hybrids add local semantics [10]. The current state of the art comes from transformer families (BERT, RoBERTa, XLNet) whose self-attention models complex syntax and semantics, consistently outperforming earlier methods [11].

Multilingual pre-training broadens coverage for low-resource languages: XLM-RoBERTa spans 100 tongues [12], yet language-specific models like PhoBERT achieve even higher scores on Vietnamese NLP benchmarks [13]. Ongoing work therefore balances accuracy with model efficiency for real-world, multilingual deployment.

B. Voice-based Emotion Classification

Speech Emotion Recognition (SER) underpins natural HCI by mapping acoustic signals to affective states. Before 2020 most systems combined handcrafted features—MFCCs, pitch, energy, speech-rate—with classical classifiers such as SVM, HMM, or GMM [14]. These pipelines were light but captured limited temporal context.

Deep-learning models soon replaced them: RNN/GRU and CNN architectures learn emotion cues directly from raw audio. An Attention-GRU with Dynamic-CBAM, for instance, attains 87% accuracy ($F1 = 0.87$) on the Vietnamese VNEMOS corpus and even supports depression screening [15].

Self-supervised pre-training marks the current frontier. Wav2Vec 2.0 and, especially, HuBERT provide context-rich speech embeddings from unlabelled data; HuBERT outperforms Wav2Vec 2.0 on IEMOCAP (79.6% vs 73.0% WA) and pushes VNEMOS to 89% accuracy across five emotions [6], [16].

Despite these gains, SER still struggles with background noise, speaker variability, and low-resource languages—issues our study tackles by pairing HuBERT with a complementary text branch.

C. Multimodal Emotion Recognition

Multimodal Emotion Recognition (MER) involves identifying human emotions by integrating data from various sources, such as text, speech, and facial expressions. This integration enhances accuracy and understanding, particularly in applications like human-computer interaction. Common fusion strategies include Early Fusion, which combines features from different modalities at the input level; Late Fusion, which processes each modality separately and merges the outputs; and Intermediate Fusion, which integrates features at intermediate layers, often using architectures like Transformers to model complex interactions [17]. Additionally, research has categorized fusion methods into data-level, feature-level, and model-level fusion, discussing challenges in multimodal data integration and suggesting future research directions to enhance scalability and accuracy [18].

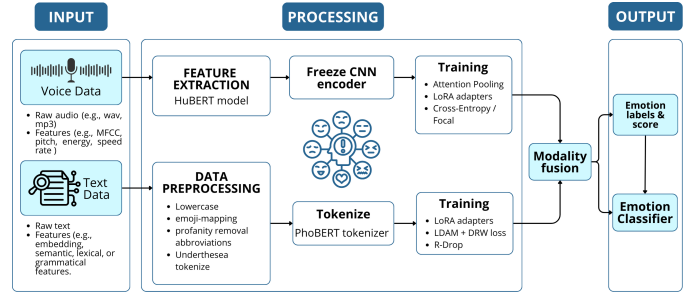


Fig. 1. End-to-end multimodal emotion-recognition pipeline integrating voice and text.

Building upon these fusion strategies, the Multimodal Transformer Augmented Fusion (MTAF) model boosts emotion recognition by combining speech and text using both feature-level and model-level fusion. It captures intra- and inter-modal interactions through Transformer-based modules, leading to strong performance on datasets like IEMOCAP and MELD, especially for emotions such as anger and sadness [19].

Overall, fusion strategies play a crucial role in enhancing MER systems. Selecting an appropriate fusion method depends on data characteristics and application goals. While current research has laid a solid foundation, challenges remain, including handling heterogeneous data, addressing data scarcity or noise, and developing models with better generalization capabilities.

Vietnamese studies still tackle emotion separately: speech works fine-tune PhoBERT-CNN or Whisper for SER but stop at the audio track, reporting strong gains on local corpora [15] while text papers rely on PhoBERT to push emovji-enhanced emotion or sentiment accuracy yet ignore the voice channel [13]. This gap suggests building a unified MER system and, as a low-cost start, keeping the two models separate but letting each utterance be classified by the modality with the higher confidence score—a strategy still largely unexplored for Vietnamese.

III. METHODOLOGY

A. System Architecture Overview

Our system, as shown in Fig. 1, follows two parallel branches that meet in a lightweight confidence-based fusion.

The voice branch processes raw 16kHz audio through a frozen HuBERT encoder. To retain its rich self-supervised representations while reducing overfitting and computation, the convolutional feature extractor is kept frozen during training. An attention-pooling layer and classification head are appended, and only these final layers are fine-tuned, ensuring efficient adaptation to the emotion task without altering pretrained weights [20].

In the text branch, utterances are lower-cased, emoji-mapped, cleaned of profanity and abbreviations, then tokenized for PhoBERT. The model is fine-tuned using LDAM loss with Deferred Re-Weighting and R-Drop regularization to address class imbalance and improve robustness. This

TABLE I
TEXT DATASET STATISTICS AND SOURCE DISTRIBUTION

Label	Number of Rows	Source Information (%)
Enjoyment	3966	Social Networks: 70% Product Reviews: 5% Dialogues: 25%
Anger	3900	
Sadness	3997	
Fear	3827	
Surprise	3736	
Neutral	3292	

strategy allows PhoBERT to first learn generalizable features before emphasizing rare classes, enhancing performance on imbalanced emotion data.

At inference, both branches return posterior probabilities. The fusion module simply selects the prediction with the higher confidence score, avoiding heavy cross-modal networks while still exploiting whichever modality is more reliable for a given input. Preliminary experiments show that this “best-of-both” rule boosts macro-F1 without extra latency, making it attractive for real-time Vietnamese emotion-aware applications.

B. Text Emotion Classifier

The text samples utilized in this study are sourced from an extended version of the UIT-VSMEC corpus [8], which has been further enriched with in-house recordings, each meticulously annotated for six distinct emotional categories. The data is collected from diverse online sources, including social media comments, personal confessions from online diaries and blogs, as well as product reviews from e-commerce platforms. The dataset is then processed through a rigorous preprocessing pipeline to ensure quality before being fed into the model for training. Table I summarizes the dataset’s statistics, showing the number of rows for each emotion label and the distribution of sources.

The preprocessing begins with the expansion of commonly used abbreviations and the removal of inappropriate language, profanity, and special symbols. Additionally, repeated characters are stripped, and Unicode characters are normalized. The data is then tokenized using the Underthesea tokenizer [24], which prepares the inputs for model training by ensuring clean and consistent text representation.

To encode the text, we choose PhoBERT-v2, a monolingual model based on the RoBERTa architecture, which has been pre-trained on approximately 140 GB of Vietnamese text (comprising 20 GB of Wikipedia data and 120 GB of OS-CAR). PhoBERT-v2 uses dynamic masking and whole-word tokenization, which allows it to capture the linguistic subtleties inherent in the Vietnamese language, while maintaining a relatively compact model size of 135 million parameters [13]. Despite having fewer parameters than XLM-RoBERTa (which contains 250 million parameters), PhoBERT-v2 consistently outperforms its counterpart, demonstrating superior performance with a +0.05% absolute gain in the VLSP sentiment analysis task and achieving up to a +3% improvement in Vietnamese named entity recognition (NER) and dependency

parsing benchmarks. This indicates stronger transferability to downstream tasks relevant to our study [12].

After data collection, the text undergoes through pre-processing to ensure consistency and quality. Emoticons are mapped to their corresponding emotion words to aid model interpretation. Abbreviations and profanities like “dm” or “vl” are removed or replaced with neutral terms to maintain dataset integrity. Additionally, special characters such as @, #, and & are stripped to reduce noise and ensure clean input for training.

After this pre-processing stage, the dataset is ready for training, where the model undergoes a structured learning process in multiple phases. Phase 1 uses the cross-entropy loss [21]:

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \log p_{\theta}(y_i | x_i) \quad (1)$$

Phase 2 switches to LDAM + DRW, adding a label-distribution-aware margin loss with deferred class reweighting [22]. For each class j , a margin is defined as:

$$\Delta_j = \frac{C}{n_j^{1/4}} \quad (2)$$

where n_j is the number of training examples in class j , and C is a hyperparameter.

The LDAM loss is formulated as:

$$\mathcal{L}_{LDAM} = -\log \left(\frac{e^{z_{y_i} - \Delta_{y_i}}}{e^{z_{y_i} - \Delta_{y_i}} + \sum_{k \neq y_i} e^{z_k}} \right) \quad (3)$$

where z_k is the logit for class k , and y_i is the ground-truth class label for input x_i .

To further mitigate class imbalance, we apply Deferred Re-Weighting (DRW):

$$w_j = \begin{cases} 1 & \text{if epoch} < T, \\ \frac{1}{n_j} & \text{if epoch} \geq T, \end{cases} \quad (4)$$

where T is the epoch threshold for enabling re-weighting.

The final combined LDAM-DRW loss is:

$$\mathcal{L}_{LDAM-DRW} = w_{y_i} \cdot \mathcal{L}_{LDAM} \quad (5)$$

Optionally, we add R-Drop regularization to reduce variance [23]:

$$\mathcal{L}_{R-Drop} = \frac{1}{|\mathcal{B}|} \sum_{x \in \mathcal{B}} \text{KL}(p_{\theta}(\cdot | x) \| p_{\theta}(\cdot | x)) \quad (6)$$

Thus the total Phase 2 loss becomes

$$\mathcal{L} = \begin{cases} \mathcal{L}_{LDAM} + \alpha \mathcal{L}_{R-Drop} & \text{if epoch} < T, \\ w_{y_i} \cdot (\mathcal{L}_{LDAM} + \alpha \mathcal{L}_{R-Drop}) & \text{if epoch} \geq T, \end{cases} \quad (7)$$

C. Voice Emotion Classifier

In this study, we fine-tune the HuBERT model for emotion recognition from speech [20]. The dataset consists of voice recordings exhibiting various emotional tones, normalized to a 16kHz sample rate to ensure high-quality audio and computational efficiency. This sample rate is common in

TABLE II
VOICE DATASET STATISTICS

Label	Number of Rows
Happy	2779
Angry	2752
Sad	2633
Fearful	2754
Surprised	2722
Neutral	2766

speech processing and aids in extracting meaningful emotional features.

We utilize the Bud500 dataset, which includes over 600,000 spoken Vietnamese audio samples [27]. Originally designed for automatic speech recognition (ASR), the dataset was augmented with emotion labels through manual annotation of approximately 20,000 clips. Table II provides detailed statistics on the final labeled dataset, highlighting the balanced distribution of samples across six emotional categories: Happy, Angry, Sad, Fearful, Surprised, and Neutral. Additionally, the dataset ensures demographic diversity, encompassing a wide range of genders, ages, and regional accents.

Data preprocessing included volume normalization to ensure consistent loudness and background noise reduction to preserve speech clarity, critical for accurate emotion recognition. Feature extraction was performed using the pretrained facebook/hubert-base-ls960 model, leveraging a CNN-based self-supervised approach to produce high-level speech embeddings [20]. During fine-tuning, the CNN encoder was frozen to avoid overfitting, allowing only the classification layers to be updated, which has been shown to reduce model complexity and overfitting, especially in small datasets [16].

The dataset was balanced across emotional categories to prevent bias and ensure robust emotion classification. We applied feature normalization to ensure consistent formatting and quality of speech features before model training. Specifically, audio clips were volume-normalized and background noise was reduced, enhancing the clarity and uniformity of the input features, which are crucial for effective emotion recognition from speech. To further optimize training, we employed early stopping and mixed precision training (fp16) to prevent overfitting and accelerate computations while maintaining model accuracy [25]. The model was trained using cross-entropy loss, and performance was evaluated using standard classification metrics, including Precision, Recall, and F1-score, to comprehensively assess the model's performance [21].

By leveraging the Bud500 dataset and applying thorough preprocessing, this study aims to improve the accuracy and generalization of HuBERT for emotion recognition in Vietnamese speech, contributing to the advancement of emotion recognition in the Vietnamese language.

D. Fusion Strategy

In this study, we propose a fusion strategy for emotion recognition from multimodal data, specifically combining textual and voice-based information. The goal of this strategy is to

leverage the strengths of both text and voice models to improve the overall performance of emotion classification by selecting the most confident prediction between the two modalities.

When only one modality is available, the corresponding model is used for emotion classification. For text-based input, we utilize a pre-trained text classification model, such as a BERT-based architecture fine-tuned for emotion recognition. In the case of voice-based input, a speech emotion recognition model, such as HuBERT, is used to classify the emotional tone of the speech.

When both text and voice data are available, we compute softmax confidence scores for each modality. The softmax function transforms the raw output logits of the models into probabilities, which represent the confidence of the model in predicting each emotion category. The softmax function is defined as:

$$P(y_i | x) = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (8)$$

Where: - z_i is the logit for the emotion class i , - $P(y_i|x)$ is the probability of class i given the input x , - The sum in the denominator ensures the probabilities across all emotion classes sum to 1.

Given the probability distribution $P(y_i | x)$ produced by each model, we define the **confidence** of a modality $m \in \{\text{text}, \text{voice}\}$ as its maximum class probability:

$$\hat{c}_m = \max_i P_m(y_i | x) \quad (9)$$

When both modalities are present, the system compares these confidence values and keeps the label associated with the larger one:

$$\text{label} = \begin{cases} \text{label}_{\text{voice}}, & \text{if } \hat{c}_{\text{voice}} > \hat{c}_{\text{text}}, \\ \text{label}_{\text{text}}, & \text{otherwise.} \end{cases} \quad (10)$$

A tie ($\hat{c}_{\text{voice}} = \hat{c}_{\text{text}}$) is rare; in such cases we default to the *voice* label. This direct comparison assumes both models output well-calibrated probabilities; if calibration differs, we first apply *temperature scaling*. Relying on the more confident modality in this way has proved effective in settings where multimodal data are common, including virtual assistants, customer-service chatbots and sentiment-analysis platforms [26].

IV. EXPERIMENTS AND RESULTS

A. Setup

For training, the dataset is split into 80% for training, 10% for validation, and 10% for testing. This division allows for effective model training while ensuring reliable evaluation. The model performance is assessed using standard classification metrics: Accuracy, Precision, Recall, and F1-score (both macro and weighted). These metrics are chosen to provide a comprehensive evaluation, particularly for imbalanced datasets, as they account for the model's ability to classify each emotion category accurately [28]. All experiments were carried out on a server running Ubuntu 22.04.5 LTS, featuring an NVIDIA

A100 SXM4 80 GB GPU (GA100), an AMD EPYC 7742 64-core CPU, and 64 GB of RAM.

B. Baseline Models

For text-based emotion classification, we begin with SVM + TF-IDF, a straightforward yet effective method for converting text into vectors, highlighting important words for classification. This approach is particularly useful for small datasets and simple tasks [29]. We also use a CNN-LSTM model with word embeddings to extract local features via CNN and capture long-term dependencies through LSTM, enabling the model to understand both syntactic and semantic structures [30]. PhoBERT-base-v2, a pre-trained Vietnamese transformer model, is employed as a baseline, fine-tuned with cross-entropy loss. It has demonstrated superior performance in Vietnamese emotion classification due to its ability to capture linguistic nuances [13]. We further improve this by combining PhoBERT with LDAM + DRW, addressing class imbalance through label-distribution-aware margins and dynamic re-weighting.

For voice-based emotion classification, we start with SVM using MFCC features, a classic method for speech recognition, followed by CNN-LSTM on spectrograms to learn spatial features and model temporal dependencies [31]. We also evaluate GRU-based RNN, which is more efficient than traditional RNNs and handles vanishing gradients effectively [32]. Finally, HuBERT-base, a self-supervised model pre-trained on large amounts of unlabeled speech data, captures high-level phonetic features and excels in emotion recognition tasks [20].

C. Results

The results presented in the Table III for text classification clearly show that PhoBERT with LDAM + DRW outperforms the baseline models, particularly SVM + TF-IDF, CNN-LSTM, and PhoBERT-base-v2. The use of LDAM + DRW effectively addresses class imbalance, leading to a noticeable improvement in accuracy and F1-scores. This suggests that fine-tuning PhoBERT with additional techniques can significantly enhance performance for emotion classification tasks in Vietnamese text.

In Table IV, for voice classification, HuBERT-base demonstrates a clear advantage over traditional models such as GRU-based RNN and CNN-LSTM. The superior performance of HuBERT can be attributed to its self-supervised pre-training on large amounts of unlabeled speech data, allowing it to capture high-level phonetic features and improve classification accuracy. These findings emphasize the effectiveness of HuBERT in handling emotion recognition from speech, as compared to older, less advanced models.

The effectiveness of the confidence-based fusion strategy was illustrated through a representative inference example using both modalities. For an input utterance (`test.wav`), the voice model predicted *Anger* with a confidence of 0.927, while the text model predicted the same emotion with lower confidence (0.710). Thus, the final classification relied on the voice modality. This example highlights the advantage of

dynamically selecting the most confident modality, effectively resolving discrepancies between textual and vocal cues, which is particularly beneficial in real-world multimodal emotion recognition tasks.

TABLE III
COMPARISON OF TEXT CLASSIFICATION RESULTS

Model	Accuracy	Precision	Recall	F1-score
SVM + TF-IDF	70.03%	70.03%	70.03%	69.95%
CNN-LSTM	62.33%	62.42%	62.33%	62.23%
PhoBERT-base-v2 (Cross-Entropy)	80.69%	80.65%	80.68%	80.65%
PhoBERT with LDAM + DRW	87.35%	87.26%	87.26%	87.28%

TABLE IV
COMPARISON OF VOICE CLASSIFICATION RESULTS

Model	Accuracy	Precision	Recall	F1-score
SVM + MFCC	28.65%	26.98%	28.65%	27.16%
CNN-LSTM on Spectrograms	60.32%	60.05%	60.32%	60.18%
GRU-based RNN	64.87%	64.55%	64.87%	64.51%
HuBERT-base	82.06%	82.86%	82.06%	81.93%

V. CONCLUSION

Combining voice and text modalities through a confidence-based fusion approach shows potential for improving emotion recognition accuracy. By incorporating PhoBERT with LDAM + DRW, we aim to address the long-tail distribution in text-based emotion classification, which may improve the recognition of underrepresented emotions. This method aligns with recent research highlighting the benefits of label-distribution-aware margins in addressing class imbalance [13], [22]. On the voice side, HuBERT-base outperforms traditional models such as GRU-based RNN and CNN-LSTM, thanks to its self-supervised learning on large-scale unlabeled speech data, which enables it to capture fine-grained phonetic features [20].

The fusion strategy proposed here is simple yet potentially effective, making it suitable for deployment in real-world applications. By choosing the prediction with the highest confidence between voice and text, the approach achieves improved performance without adding significant complexity, which could make it suitable for resource-constrained environments [33].

Future directions include investigating more sophisticated fusion techniques, such as attention-based models, to better capture the interactions between voice and text features [34]. Additionally, testing the model on multilingual emotion recognition datasets will assess its generalization across languages and cultures, as multilingual emotion recognition continues to gain importance in recent studies [34]. Scaling up the model with larger multimodal emotion datasets will further improve its robustness and real-world applicability [33].

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